

KRUPA N

EMBEDDED SOFTWARE DEVELOPER

CONTACT

+91 9632169739
krupa.varun5@gmail.com
www.linkedin.com/in/krupa-n-805836109
Bengaluru, India

EDUCATION

Bachelor of Engineering
Electronics and
Communication
Gopalan College of Engineering
and Management
Bengaluru, India
2015

SKILLS

// PROGRAMMING

C
C++
CAPL
Python

// TECHNICAL

ECU Software Development
UDS
DCM
DEM
COM Stack
NVM Management
Bootloader

SUMMARY

Embedded Software Engineer with 10 years of experience, predominantly in automotive systems complemented by aerospace exposure. Proficient in C and C++ programming and testing methodologies with proven expertise in ASW, BSW, MCAL layers development. Strong expertise across ASPICE SWE.1–SWE.4 lifecycle including requirements, design, implementation, and verification in eHeater, eTurbo and Fuel Injection projects. Skilled in debugging, root-cause analysis, and delivering production-quality software aligned with ISO 26262 and MISRA-C standards.

EXPERIENCE

SENIOR SOFTWARE ENGINEER
Borgwarner, Bengaluru

01/2019 – 02/2026

- Began as Unit Testing Lead, driving task planning, assignment, and test case development and reviews, while actively contributing to process definition and ensuring adherence to ASPICE standards.
- Spearheaded the first successful on-target unit testing implementation for the eTurbo product using Tessy on SPC58EC microcontroller with GHS compiler and Lauterbach TRACE32 debugger.
- Progressed into software development responsibilities delivering end-to-end contributions across SW1 requirement writing, SW2 architecture design and SW3 implementation.
- Optimized legacy software across UDS diagnostics and NVM modules, resolving critical defects and improving system stability and performance.
- Reduced ECU flashing time by ~80% (10 min -> 2 min) using dual-baud rate strategy addressing challenges in NXP SBC configuration over UART and SPI.
- Implemented immediate NVM write strategy for error counters, ensuring data persistence during improper shutdown scenarios
- Developed complex diagnostic functionalities for eHeater systems, including derating and power control strategies aligned with system safety and performance requirements.
- Instrumented application code to monitor CPU load metrics, enabling performance analysis and optimization of real-time system behaviour.
- Configured peripherals including GPIO, ADC, PWM, and communication interfaces, mapping HSI signals using PSoC Creator for Porsche 10kW dual-track eHeater program.
- Implemented interlock safety functionality with diagnostic support for eHeater systems, and optimized PWM frequency to resolve electromechanical issues, enhancing system reliability and operational stability.
- Enhanced power control, signal acquisition and heater application modules to improve system efficiency and functional performance.
- Revamped memory layout and optimized NVM configuration to improve memory utilization, data handling efficiency, and system reliability.
- Implemented flashing solution using VFlash and ODX, integrating third-party bootloader for reliable ECU reprogramming and diagnostics support.

// AUTOSAR & STANDARDS

MCAL/BSW Integration
ASPACE (SWE.1 – SWE.4)
ISO 26262
MISRA-C
DO-178B

// PROTOCOLS

CAN
UART
LIN
SPI

// TOOLS

CANoe
EB Tresos
Visual Studio
PSoC Creator
VFlash
ODX Flash
TRACE32 (Lauterbach)
Ozone
Tessy
RTRT
Polyspace
QAC

// MICROCONTROLLERS

S32K118 (NXP)
Cypress (Now Infineon)
SPC58EC (ST)

// VERSION CONTROL & CI/CD

GIT
BITBUCKET
JENKINS

EXPERIENCE continued

- Configured CAN MCAL and generated BSW modules using EB Tresos for NXP S32K118 microcontroller, enabling integration of AUTOSAR-compliant CAN stack within a customized (non-AUTOSAR) architecture for commercial vehicle projects.
- Led code Integration, smoke testing and coordinated internal releases.
- Customized Python scripts for automated binary generation.
- Conducted technical reviews and audits, ensuring adherence to MISRA C, ISO 26262, and Automotive SPICE standards.
- Acted as mentor and first-line problem solver, ensuring team productivity and code quality through peer reviews.
- Key contributor to the 'Be the Best' initiative, driving ideas that boosted team productivity and efficiency.
- Contributed abroad through onsite training and cross-functional collaboration, shaping processes and resolving key project challenges.

SENIOR SOFTWARE ENGINEER GRADE 1
AK Aerotek, Bengaluru

02/2017 – 01/2019

- Understating SW requirement and implement code.
- Follow Misra-C 2004 Coding Rules during the Coding Phase.
- Perform QAC analysis to maintain good quality of software.
- Developed low-level requirements and reviewed test cases.
- SW requirements are validated on test bench.
- Performing the Unit test using RTRT.
- Responsible for peer reviews to adhere with software quality.

CONTINGENT WORKER
Aeronautical Development Agency, Bengaluru

09/2015 – 02/2017

- Coordinated with clients and technical teams for smooth communication and issue resolution.
- Developed manual test cases for simulation testing and conducted peer reviews.
- Tracked defects, analysed requirements, and validated Brake Management System using MATLAB.
- Performed periodic maintenance checks on hardware/software for reliability.

REWARDS & ACHIEVEMENTS

- Customer Collaboration Award — Recognized for collaboration on the eTurbo project and succeeding in test-on-target execution for the first time.
- Spark Awards (Quarterly) — Frequently awarded for performance excellence and innovative "Be the Best" ideas
- Hercules Task Award — Honoured for critical contribution to external bootloader integration for the eHeater product and achieving successful flashing within a short span of time.

LANGUAGES

- English
- Kannada
- Tamil
- Telugu
- Hindi